

Data Modernisation & Data Processing

HCQO

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Overview

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Why Data Modernise?

- Efficiency gain: automation (semi-automate, manual), replicability
- Transparency: collaboration, documentation
- Increase in data quality: via validation tools

Takeaway

The goal is to improve data quality by making better use of efficient, purpose-built tools.

Data Processing



- DataConstruction: inputs from delegates, outputs to DataBase
- Includes:
 - Data Cleaning
 - Data Validation: HCQO_Validation_2024 (We want to minimize data cleaning)
 - Data Uploading: ResultsFormation

Project Architecture

```
DataConstruction
├── DataCleaning
│   ├── Code
│   ├── Input
│   ├── Temp
│   ├── Notes
│   ├── README.txt
│   └── Output
├── HCQO_Validation_2024
│   ├── classes
│   └── functions
├── ResultsFormation
├── Check
└── README.txt
```

Transition of Work

- Good organisation of calculation, validation codes based on different classes.
- Experience from 2025 data collection: first use of validation code
- Finalisation of `HCQ0_Validation_2024`: limited delegate adoption, needed debugging and code cleanliness

Enhancements Since 2025 May

- Expansion of developments
- Data Cleaning: compiling data by year, indicator groups
- Data Validation: updates to rates calculation (DEN 0)
- Data Uploading: reformat data structure
- Testing workflow: Check folder

Technical Overview

- Project architecture
- Running the full process
- README documentation

Next Steps

- Finalisation of HCQ0_Validation_2026
- Scope of work: systematic tool
- Example: Y_GE166 → Y_GE16 No; invalid entries flagged as errors
- Project architecture: ex) hosted on Dropbox

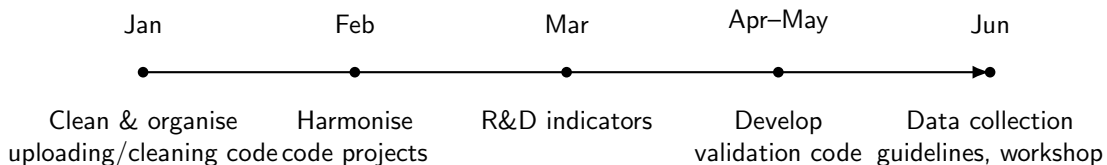
Key Point

Delegates will be responsible for correcting their input data (Enforcing Systematic Tools).

Other Related Modernisation Work

- HAG
- VISED

Tentative Timeline



Long-Term Perspectives

- Harmonize Calculation Codes
- HAG: repeated generation of similar graphs using updated datasets
- Modernise analysis: semi-automate repetitive visualizations, and their underlying data manipulations

Discussion

- Direction: delegate use vs centralized approach
- Questions